

Equations of equilibrium of concurrent force system in a plane

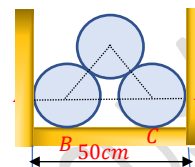
We have seen that the resultant of a system of concurrent coplanar forces can be found out using the equation, $R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$.

If the force system is in equilibrium, then the resultant of the force system should be zero. This is possible only if $\Sigma F_x = 0$ and $\Sigma F_y = 0$. Hence, the equations of equilibrium for concurrent forces in a plane are,

$$\Sigma F_x = 0 \text{ and } \Sigma F_y = 0$$

Example 1

Three balls of radius $r = 10 \text{ cm}$ and weight $W = 100 \text{ N}$ each resting on a trough as shown. Determine the support reactions at A, B, C, and D.



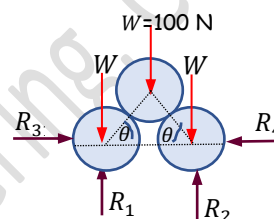
Consider the freebody diagram of the whole system

$$\Sigma F_x = 0 \gg R_3 - R_4 = 0 \gg R_3 = R_4$$

$$\Sigma F_y = 0 \gg R_1 + R_2 - 100 - 100 - 100 = 0 \gg R_1 + R_2 = 300$$

From the geometry,

$$10 + 20 \cos \theta + 20 \cos \theta + 10 = 50 \gg \cos \theta = \frac{3}{4} \gg \theta = 41.4^\circ$$



Consider the free-body diagram of ball 1

$$\Sigma F_x = 0 \gg R \cos 41.4^\circ - R' \cos 41.4^\circ = 0 \gg R = R'$$

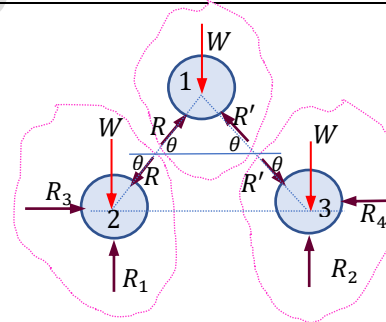
$$\Sigma F_y = 0 \gg R \sin \theta + R' \sin \theta = 100 \gg R = \frac{100}{2 \times \sin 41.4} = 75.6 \text{ N}$$

Equilibrium of ball 2

$$\Sigma F_x = 0 \gg R_3 - R \cos 41.4^\circ = 0 \gg R_3 = 56.7 \text{ N}$$

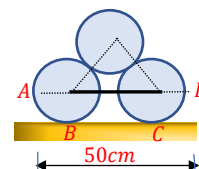
$$\Sigma F_y = 0 \gg R_1 - R \sin 41.4^\circ - 100 = 0 \gg R_1 = 150 \text{ N}$$

$$R_2 = 150 \text{ N} \ll R_4 = 56.7 \text{ N}$$



Example 2

Three balls of radius $r = 10 \text{ cm}$ and weight $W = 100 \text{ N}$ each resting on a smooth floor. Determine the support reactions at A and B and the tension if a string keeps the balls from moving.

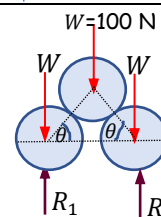


Consider the freebody diagram of the whole system

$$\Sigma F_y = 0 \gg R_1 + R_2 - 100 - 100 - 100 = 0 \gg R_1 + R_2 = 300$$

From the geometry,

$$10 + 20 \cos \theta + 20 \cos \theta + 10 = 50 \gg \cos \theta = \frac{3}{4} \gg \theta = 41.4^\circ$$



Consider the free-body diagram of ball 1

$$\Sigma F_x = 0 \gg R \cos 41.4^\circ - R' \cos 41.4^\circ = 0 \gg R = R'$$

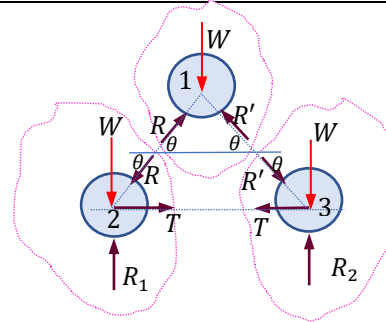
$$\Sigma F_y = 0 \gg R \sin \theta + R' \sin \theta = 100 \gg R = \frac{100}{2 \times \sin 41.4} = 75.6 \text{ N}$$

Equilibrium of ball 2

$$\Sigma F_x = 0 \gg T - R \cos 41.4^\circ = 0 \gg T = 56.7 \text{ N}$$

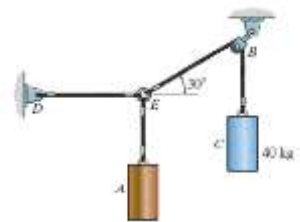
$$\Sigma F_y = 0 \gg R_1 - R \sin 41.4^\circ - 100 = 0 \gg R_1 = 150 \text{ N}$$

$$R_2 = 150 \text{ N}$$



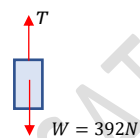
Example 3

Determine the weight of the cylinder A which will keep the system in equilibrium. Neglect the friction in the pulley at B.

**Free body diagram of the cylinder C**

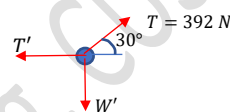
$$\Sigma F_y = 0 \gg T - W = 0 \gg T = 392 \text{ N}$$

Since, the friction in the pulley B is neglected, the tension in the string CBE will be the same

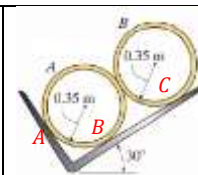
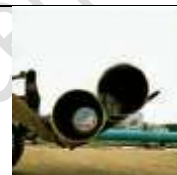
**Free-body diagram of E**

$$\Sigma F_x = 0 \gg T \cos 30 - T' = 0 \gg T' = 339.5 \text{ N}$$

$$\Sigma F_y = 0 \gg T \sin 30 - W' = 0 \gg W' = 196 \text{ N}$$

**Exercise 1.**

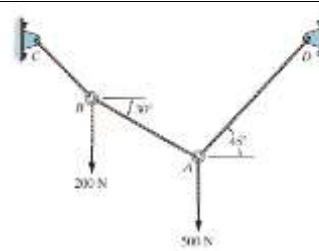
Two smooth pipes, each having a mass of 300 kg, are supported by the forked tines of a tractor as shown. Determine the reactions at the supports A, B, and C.

**Exercise 2.**

Determine the tension in the cables AB and BC if the cylinder weighs 600 N.

**Exercise 3.**

Determine the tension in the cables AB, BC and AD.

**Exercise 4**

If each cable can withstand a maximum load of 500 N, what is the maximum possible weight of the suspended cylinder?

**Reference**

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

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